

Highlighting principles:

Analytical variables(terms) are in bold

Version controls are in red and bold

Variables names (in code) are in blue

1. Data Cleaning

- a. selecting variables regarding the research focus
- This dataset is a preliminary study of housing ownership across census metropolitan areas across Canada.
- **Weights are not used in this version of analysis**
- Since people with a mortgage are entitled with housing tenure (at least during the period of investigation), the analytical dataset should be separated into two independent dataset with one using tenure as the outcome variable and the other using mortgage as the outcome variable.
 - First, filter samples where `tenure=1` into a separate dataset and construct a separate dataset `df_mortgage`; and then generated another new dataset named as `df_tenure`
 - Secondly, list-wise delete observations with missing values in outcome variables and other independent variables, separately in these two datasets
 - Then proceed to analysis

2. Definition of the analytic sample

- a. sample selection: all observations aged 15 and older from the 2016 Canada census PUMF (Public Use Micro File) data
- b. variable selection, recode, and rename

Age

Variable name: `AGEGRP`

Original coding: Age groups, with a 5-year interval. Records valued as 88 means not available

Recoded as `agegep_5`: age with a 5-year interval

Marital status

Variable name: `MarStH`

Original coding:

Value	Label	Records	Weighted
1	Never legally married (and not living common law)	375,558	13,909,511.5
2	Legally married (and not separated)	356,994	13,221,911.0
3	Living common law	94,795	3,511,043.0

4	Separated (and not living common law)	18,567	687,667.8
5	Divorced (and not living common law)	47,488	1,758,832.4
6	Widowed (and not living common law)	37,019	1,371,098.3

Recoded as [married](#):

Sex

Variable name: [Sex](#)

Original coding:

Value	Label	Records	Weighted
1	Female	472,129	17,486,294.9
2	Male	458,292	16,973,769.1

Recoded as: [female](#)

Income

Variable name: [Wages](#)

Def: Personal incomes from certain sources; Population aged 15 years and over in private households

Original coding: continuous variable. The value 88,888,888 stands for not available. The value 99,999,999 stands for not applicable and is applied to all persons aged less than 15 years. Otherwise, this variable could be positive, negative or zero and is a rounded value of the amount received by the individual in 2015. Values that would have been rounded to zero have been replaced by 1. In some cases, high values have been top coded and low values have been bottom coded in this file.

Recoded as: [income_q](#)

Records valued as 88,888,888 or 99,999,999 are recoded as missing values. The rest valid cases are coded into a three-level income group based on the quantile distribution: those below 25% are coded as 1, between 25% and 75% are coded as 2, and those higher than 75% are coded as 3.

Presence of mortgage

Variable name: [PresMortG](#)

Def: Refers to whether an owner household makes regular mortgage or loan payments for their dwelling; Persons in private households in owner-occupied private dwellings.

This variable is reported for private households in owner-occupied private dwellings.

Original coding:

Value	Label	Records	Weighted
0	Without mortgage	218,357	8,087,365.9
1	With mortgage	453,726	16,804,561.3
8	Not available	4,849	179,591.3
9	Not applicable	253,489	9,388,545.5

Recoded as: [mortgage](#)

Housing Tenure

Variable name: [Tenur](#)

Def: Refers to whether the household owns or rents their private dwelling. The private dwelling may be situated on rented or leased land or be part of a condominium.

Original coding:

Value	Label	Records	Weighted
1	Owned by a member of the household	67,083	24,891,927.2
2	Rented or Band housing	253,489	9,388,545.5
8	Not available	4,849	179,591.3

Recoded as: [tenure](#)

Education

Variable name: [HDGREE](#)

Def: This variable refers to whether a person attended, either full-time or part-time, any accredited educational institution or program at any time during the nine-month period between September 2015 and May 10, 2016. The person may have attended more than one educational institution or have been enrolled in more than one program; Population aged 15 years and over in private households

Original coding:

Value	Label	Records	Weighted
1	Did not attend school	647,718	23,989,647.6
2	Elementary or secondary school	39,978	1,480,657.7
3	Technical or trade school, community college or CEGEP	38,389	1,421,821.8
4	University	41,907	1,552,112.8
5	Multiple responses	1,452	53,778.4
8	Not available	3,873	143,444.9
9	Not applicable	157,104	5,818,600.8

Recoded as: [edu_college](#)

Geographical location

Variable name: [PR](#)

Def: Province or territory of current residence. Yukon, the Northwest Territories and Nunavut are grouped under the term 'Northern Canada'

Original coding:

Value	Label	Records	Weighted
10	Newfoundland and Labrador	13,830	512,251.0
11	Prince Edward Island	3,763	139,686.0
12	Nova Scotia	24,528	908,340.0
13	New Brunswick	19,720	730,708.0
24	Quebec	215,041	7,965,452.0
35	Ontario	357,536	13,242,160.0
46	Manitoba	33,521	1,240,699.0
47	Saskatchewan	28,894	1,070,558.0
48	Alberta	107,460	3,978,146.0
59	British Columbia	123,109	4,560,238.0
70	Northern Canada	3,019	111,826.0

Recoded as: [province](#)

c. Notes for data recoding & data cleaning

Since the variable, “Presence of Mortgage” limit respondents to 15+ years old, observations are then limited to respondents who are 15 and older.

During the recoding process, new variables are generated, and original ones remain unchanged.

3. Descriptive analysis & summary of variables used in the analysis

- Descriptive analysis is based on the recoding of the original dataset, without cleaning missing values.
- All variables are transformed into factors

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
1	agegrp_5 [factor]	1. 15–20 2. 21–25 3. 26–30 4. 31–35 5. 36–40 6. 41–45 7. 46–50 8. 51–55 9. 56–60 10. 61–65 [5 others]	53406 (7.0%) 59601 (7.8%) 60644 (7.9%) 62180 (8.1%) 60799 (8.0%) 59706 (7.8%) 62484 (8.2%) 71589 (9.4%) 69829 (9.1%) 59991 (7.8%) 144389 (18.9) %		764618 (98.8%)	9139 (1.2%)
2	marr [factor]	1. single 2. married or common law	321968 (41.6) % 451789 (58.4) %		773757 (100.0%)	0 (0.0%)
3	female [factor]	1. male 2. female	395723 (51.1) % 378034 (48.9) %		773757 (100.0%)	0 (0.0%)

4	income_q [factor]	1. bottom 25% 2. middle 50% 3. top 25%	125608 (25.0) % 251214 (50.0) % 125607 (25.0) %		502429 (64.9%)	271328 (35.1%)
5	mortgage [factor]	1. without mortgage 2. with mortgage	205078 (36.6) % 355260 (63.4) %		560338 (72.4%)	213419 (27.6%)
6	tenure [factor]	1. Non-ownership 2. Ownership	208670 (27.1) % 560338 (72.9) %		769008 (99.4%)	4749 (0.6%)
7	edu_college [factor]	1. below college 2. college or above	386923 (50.6) % 377763 (49.4) %		764686 (98.8%)	9071 (1.2%)

8	province				773757	0
	[factor]	1. Atlantic provinces	52707 (6.8%)		(100.0%)	(0.0%)
		2. Central Canada	477285 (61.7%)			
		3. Prairies	%			
		4. British Columbia	136984 (17.7%)			
		5. Northern Canada	%			
			104460 (13.5%)			
			%			
			2321 (0.3%)			

4. Model analysis

- After list-wise cleaning missing values
- Separately construct two logistic regression model on [df_tenure](#) and [df_mortgage](#)
- graph coefficient plot

